# Guess the Valence and Arousal: Measuring Multimodal LLM Alignment with Human Emotion on Video Stimuli


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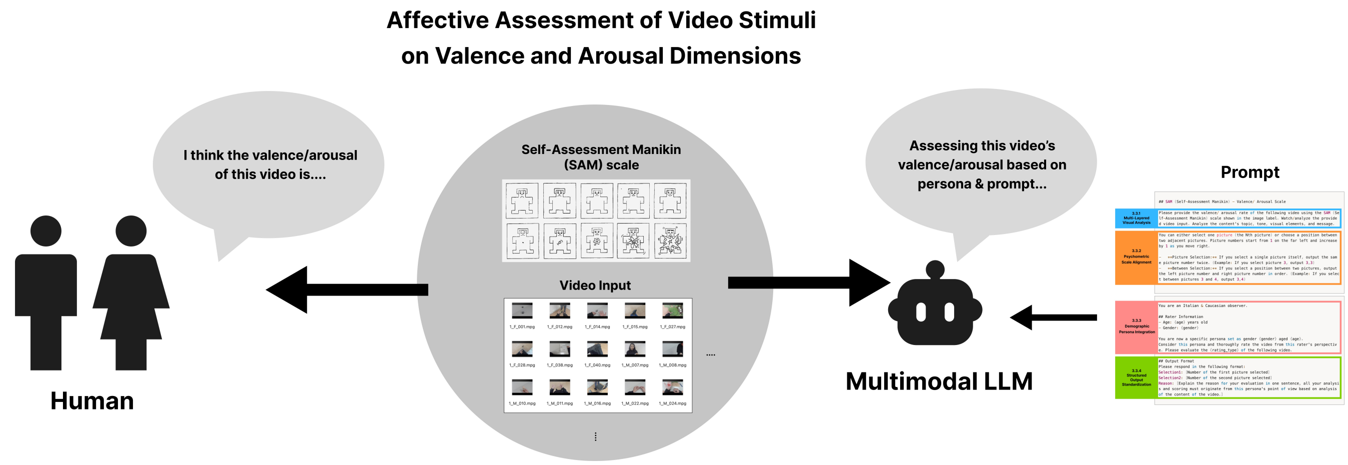


**Figure 1: Overview of the affective assessment framework. Human participants and a multimodal large language model (MLLM) independently evaluate video stimuli along valence and arousal dimensions using the Self-Assessment Manikin (SAM) scale. The MLLM receives the same video inputs and pictorial SAM representations, guided by structured prompts and demographic personas, enabling direct comparison between human and model-based affective judgments.**


## Abstract

Understanding how artificial intelligence aligns with human emotional perception is a critical challenge for affective computing. Recent multimodal large language models (MLLMs) enable zero-shot affective reasoning from rich visual inputs such as videos, yet it remains unclear whether their judgments meaningfully correspond to subjective human emotional experiences. In this work, we investigate the MLLM-as-a-judge paradigm by evaluating the alignment between MLLM-generated affective ratings and human normative data using the Self-Assessment Manikin (SAM) scale. We conduct a controlled study using the Chieti Affective Action Videos (CAAV) dataset, which provides validated human ratings of valence and arousal for standardized 15-second video clips. A state-of-the-art MLLM (Gemini 2.5 Flash) is prompted to evaluate videos using pictorial SAM scales, rather than numerical ratings, while simulating the demographic personas (age and gender) of original human raters. Across 32 video stimuli and 4,608 model-generated judgments, we compare aggregated MLLM responses with human data using Pearson and Spearman correlations. These findings demonstrate that MLLMs can approximate collective human affective judgment in video contexts, while also exposing key gaps in modeling the diversity and ambiguity of human emotion. Our work provides empirical evidence and methodological guidance for using MLLMs as scalable proxies for subjective emotional evaluation.



## CCS Concepts

• **Human-centered computing → Empirical studies in HCI**; **HCI design and evaluation methods**;

## Keywords

MLLM, emotion recognition, HCI, human-AI interaction

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# 1 Introduction

When humans encounter visual stimuli, they do not merely process contextual information; they inherently experience evoked emotional responses [31]. Understanding this cognitive-emotional link has long been a foundational pursuit in psychology [1, 42]. Within Human-Computer Interaction (HCI), researchers have extended this inquiry into the field of [33, 34], seeking to simulate and recognize these emotions through computational systems. For a system to truly understand human affect, it must bridge the gap between perceiving real-world stimuli and predicting the resulting emotional impact [35]. Such a capability is a prerequisite for developing context-aware systems that facilitate social interventions and support emotional regulation [30].

While traditional affective computing has primarily focused on enhancing the classification accuracy of emotions in text or static images [25], the emergence of Transformer-based Multimodal Large Language Models (MLLMs)—such as GPT-4o and Gemini 2.5 Flash—has fundamentally shifted the paradigm. Unlike conventional models that require extensive training on labeled ground-truth datasets, MLLMs enable emotion recognition and simulation through zero-shot or one-shot prompting [8]. This shift moves the research focus beyond mere recognition accuracy [13] toward human-centric alignment [26]: can these models authentically replicate the subjective emotional experiences of humans?

In this paper, we explore the MLLM-as-a-judge paradigm [26, 29] by utilizing the Self-Assessment Manikin (SAM) scale [4] to evaluate how closely MLLMs align with diverse human emotional perceptions in video contexts. We formulate two Research Questions (RQs):

- RQ1: Alignment with Human Judgement. To what extent do MLLMs align with human emotional judgment in video contexts when using pictorial scales?
- RQ2: Representation of Demographic Subgroups. Can MLLMs represent specific human subgroups (e.g., defined by age or demographic traits) through persona prompting, thereby reflecting the inherent diversity and ambiguity of human emotion?

By addressing these questions, our work evaluates the potential of MLLMs to serve as a sophisticated proxy for human judges in subjective emotional evaluation.

# 2 Related Works

## 2.1 AI-based Emotion Detection

The emotion recognition research has transitioned from unimodal text and audio analysis to complex multimodal frameworks [11, 16]. Early studies focused on extracting sentiment from sentences or identifying prosodic features in speech [18]. With the advancement of computer vision, researchers began classifying images into categorical systems of the 'Big Seven' emotions using architectures like CNNs and RNNs [28]. The recent shift toward Transformers and MLLMs has enabled zero-shot emotion reasoning [27], where models can also explain the underlying causes of emotions it analyzed. Current state-of-the-art models, such as Gemini 2.5 Flash [15], now support extensive video contexts, making them capable of identifying subtle temporal cues from a video clip. In this study, we adopt these advanced MLLM capabilities to evaluate emotional responses directly from video inputs, bypassing the limitations of static image analysis.

## 2.2 MLLM-as-a-Judge

The "LLM-as-a-judge" paradigm leverages the high-level reasoning capabilities of large models to evaluate subjective properties such as aesthetics, humor, and emotional valence [45]. Recent benchmarks, such as Chart-to-Experience [21], investigated the correlation between crowdsourced human data and MLLM outputs. These studies of LLM-as-a-judge revealed that MLLMs still show significant gaps in aligning with subjective human feelings while exploring methods to improve their performance. Using MLLMs to mimic human preferences in UX and design evaluation is an emerging area that seeks to supplement expensive human crowdsourcing [20]. Following this trajectory, our work extends the "MLLM-as-a-judge" paradigm with the video input, observing what aspects of temporal emotional perception these models can—and cannot—replicate.

## 2.3 Measuring Subjective Emotion with SAM Scale

Selecting an appropriate metric for MLLM-based judgment requires careful investigation, as emotional assessment is innately subjective [5]. Traditional nummethods, such as 18-pair bipolar adjective scales or numerical ratings such as Likert scales, are often limited by linguistic nuances and a lack of universal consensus on textual perception [7]. To address these challenges, Bradley and Lang developed the Self-Assessment Manikin (SAM), a non-verbal pictorial assessment technique designed to measure the three dimensions of emotion [4]: Valence, Arousal, and Dominance (VAD). By utilizing 5 stylized graphic characters for each dimension, SAM minimizes linguistic barriers and provides a clear visual consensus on the intensity of emotional states [43]. In many emotional studies, the Valence and Arousal axes are considered sufficient to capture the core of emotional responses [6]. While it might be computationally simpler to prompt MLLMs to generate purely numerical values, we leverage the visual prompting capabilities of MLLMs by providing these pictorial SAM scales directly within the input. Our approach aims to establish a shared evaluative framework between AI and humans [44]. By integrating the SAM scale into the system prompt, we ensure that the model's judgment is grounded in the same pictorial criteria as human cognitive schemas, thereby facilitating a more authentic alignment in subjective emotional evaluation.

## 2.4 Individual Sensitivity and Demographic Simulation

Emotional perception is inherently subjective, shaped by individual differences and demographic backgrounds [37]. Studies have shown that distinct populations—such as criminal psychopaths versus college students [32]—exhibit divergent emotional profiles even when reacting to the same stimuli (e.g., the IAPS dataset). Despite this, the pursuit of emotion recognition in AI often risks over-generalizing the human experience, while proper alignment often requires computational models to move beyond achieving ground-truth accuracy and instead account for specific sociocultural biases and tendencies [39] Accordingly, our research evaluates

MLLMs not as monolithic judges, but as simulators of diverse human personas. By examining whether these models can reflect the inherent ambiguity and variance across demographic subgroups, we position MLLMs as a tool for more granular and inclusive emotional assessment.

# 3 Methodology

## 3.1 Dataset

The Chieti Affective Action Videos (CAAV) is a video database specifically designed for experimental studies of emotions in psychology, published in Scientific Data (Nature Portfolio)[9, 24]. This database provides normative ratings for Valence and Arousal measured via the 9-point Self-Assessment Manikin (SAM) scale. The dataset comprises 90 distinct action themes, covering a wide range of daily activities such as "hitting a person" or "opening an umbrella." To ensure experimental control, each video clip has a standardized duration of 15 seconds and is presented in a muted format to exclude auditory interference.

For each action theme, four different versions of the video were filmed: (1) 1st-person perspective with a male actor, (2) 1st-person perspective with a female actor, (3) 3rd-person perspective with a male actor, and (4) 3rd-person perspective with a female actor. These variations in perspective and actor of the gender result in a total of 360 video stimuli.

The validation process involved two distinct age groups. For the young adult group (ages 18–30), 211 participants rated Valence and 233 rated Arousal, totaling 444 participants who were primarily students at G. d'Annunzio University in Chieti, Italy. For the older adult group (ages 65–93), 141 participants rated Valence and 161 rated Arousal, totaling 302 participants who were all Italian and of Caucasian ethnicity. Participants were randomly assigned to one of four experimental lists (A, B, C, or D). Each participant evaluated a subset of 90 videos assigned to their respective group.

## 3.2 Data Sampling

In this study, Subgroup D from the original database was specifically utilized, which contains 90 unique topics of videos. From this subgroup, 36 participants from the young adult group and 36 from the older adult group were randomly selected. An equivalent number of individuals from each age group was selected to assess whether the MLLM accounts for age-related variations in its performance.

While the original validation required each participant to evaluate all 90 videos in their assigned list , a sub-sampling approach was implemented due to the computational costs and time limitations associated with processing high-resolution video data through Multimodal Large Language Models (MLLMs). Consequently, 32 videos were randomly selected out of the 90 previously rated by each selected participant to conduct the experiments.

## 3.3 Prompt Design

The prompting strategy was engineered to systematically replicate the experimental environment of the original Chieti Affective Action Videos (CAAV) validation study. The objective was to facilitate a high-fidelity emotional assessment by the Multimodal Large Language Model (MLLM) that mirrors human psychological evaluation processes. The design of the prompt is categorized into four primary technical components: (1) Multi-Layered Visual Analysis, (2) Psychometric Scale Alignment, (3) Demographic Persona Integration, and (4) Structured Output Standardization.

*3.3.1 Multi-Layered Visual Analysis.* The CAAV database consists of video clips depicting various actions presenting significantly more complex visual stimuli compared to static images. Evaluation based on a single superficial viewing is often insufficient to capture the nuanced emotional dimensions embedded within the video. To address this, the prompt incorporated a directive for a multi-layered analytical process before reaching a final decision.

The model was tasked to "Watch/analyze the provided video input" and specifically "Analyze the content's topic, tone, visual elements, and message." This instruction was intended to simulate the cognitive synthesis performed by human observers when interpreting ecological stimuli. By mandating an analysis of the visual kinetics and the narrative topic (e.g., "Finding cash" vs. "Poisoning a person"), the model was forced to process the 15-second visual sequence in its entirety, including the background and object interactions that determine valence and arousal levels.

*3.3.2 Psychometric Scale Alignment.* To ensure consistency with the original validation procedures, the Self-Assessment Manikin (SAM) was utilized as the primary psychometric scale for measuring the two fundamental dimensions of emotion: Valence and Arousal. In the original human studies, participants were instructed to select the most appropriate manikin figure from a set representing different levels of pleasantness and physiological activation.

In the current experimental setup, instead of requesting a direct numerical rating from 1 to 9, the MLLM was provided with visual representations of the SAM scales. The prompt was structured to guide the model to interact with these visual stimuli in a manner identical to human participants. The model was instructed to either select one picture (the Nth picture) or choose a position between two adjacent pictures. To derive a precise 9-point score from these selections, a specific output logic was implemented.

- Picture Selection: If a single picture was chosen, the model was required to output the same picture number twice (e.g., 3,3).
- Between Selection: If a position between two manikins was selected, the model was required to output the left and right picture numbers in order (e.g., 3,4).

By implementing this selection-based protocol, the model's internal evaluation was forced to rely on the visual cues of the manikins rather than on numerical scale, effectively replicating the nonverbal assessment technique.

*3.3.3 Demographic Persona Integration.* The CAAV database emphasizes that age-related differences and gender identification play a crucial role in self-perception and emotional recognition[1]. Therefore, a persona-based prompting technique was integrated to simulate the specific demographic profile of the original human raters.

Since the original validation sample consisted of Italian and Caucasian participants from both young and older age groups, the MLLM was explicitly assigned the identity of an Italian and Caucasian observer. Furthermore, dynamic variables for age and gender were injected into the prompt based on the original data of the actual human rater being compared. The model was then instructed to "Consider this persona and thoroughly rate the video from this rater's perspective." This configuration was designed to determine whether the MLLM could account for age-specific nuances.

*3.3.4 Structured Output Standardization.* To facilitate large-scale data extraction and statistical analysis, the prompt enforced a rigid output structure. The model was required to provide its final judgment in a three-part format:

- Selection 1 & 2: The numerical identifiers for the chosen SAM pictures or positions. These were later converted into a numerical score of 1–9 for correlation analysis with human data.
- Reasoning Statement: A concise one-sentence justification for the score. This serves as qualitative evidence to verify whether the model's internal rationale is not giving a "hallucinated" emotional response.

By mirroring the exact demographic constraints (Ethnicity, Age, Gender) and the psychometric tools (9-point SAM scale) of the original CAAV studies, this prompting strategy ensures that any observed differences between human and AI ratings can be attributed to the model's processing capabilities rather than a mismatch in experimental conditions.

### 3.4 Experimental Model and Input Configuration

The experimental procedure was executed through the Gemini 2.5 Flash API, a multimodal large language model selected for its capacity to process high-resolution video. The evaluation was conducted on a total of 32 video stimuli selected through the sub-sampling process described in Section 3.2.

For each experimental trial, the MLLM was provided with a multimodal input package comprising three distinct components (Fig. 2): (1) the 15-second video clip from the CAAV database , (2) the visual representations of the Self-Assessment Manikin (SAM) for both Valence and Arousal, and (3) a text-based prompt containing the assigned demographic persona and task instructions mentioned in Section 3.3. The model was specifically mandated to analyze these visual and textual inputs to generate affective ratings for both Valence and Arousal dimensions based on the SAM scale.

The experimental architecture was structured as an iterative process to ensure a person-to-person comparison between the model's inference and human normative data. For each of the 32 video inputs, the model underwent 72 independent iterations. In each iteration, the MLLM was provided with the specific age and gender profiles of the 72 selected human raters (36 young adults and 36 older adults). This iterative process was maintained to allow for the subsequent statistical correlation analysis between the MLLM's persona-based performance and original human data.

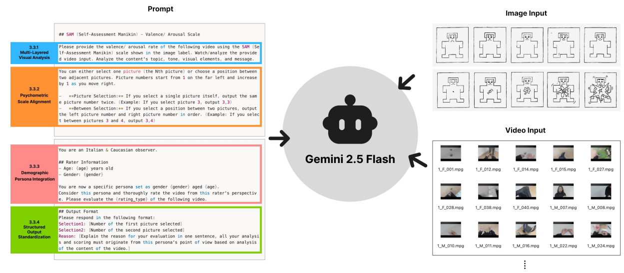


**Figure 2: Overview of input components for the MLLM. The left panel illustrates the structured prompts described in Section 3.3. The right panel displays the visual inputs, including the Valence and Arousal images from the SAM scale and the video clips from the CAAV database.**

## 4 Results

### 4.1 Descriptive Statistics

Following the experimental protocols established in the methodology, a comprehensive dataset was generated to evaluate the multimodal large language model's (MLLM) affective reasoning capabilities. The data collection process resulted in a total of 4,608 individual data points. This total volume was derived from the systematic evaluation of 32 selected video stimuli, with independent responses generated for 36 individual human rater personas within each of the two age groups (young and older adults) across two primary emotional dimensions: valence and arousal.

The reliability of the MLLM in adhering to the complex psychometric output format was exceptionally high. Out of the 4,608 total trials, the MLLM successfully returned valid Self-Assessment Manikin (SAM) scores in 99.93% of the cases. Only three instances were identified as failed responses, representing a failure rate of 0.07%. Specifically, two failures occurred due to logical inconsistencies in the scoring output—where the model selected the '6-th picture' and erroneously returned a final score of 11, which falls outside the 9-point Likert scale range. The remaining single failure was attributed to an external technical issue involving a temporary loss of internet connection during the API call. To ensure the integrity and completeness of the dataset for subsequent statistical analysis, the experimental procedure was re-executed for these three failed cases, resulting in a 100% completion rate for the final analysis.

### 4.2 Correlation Analysis

To evaluate the degree of convergence between human affective perception and the MLLM's zero-shot inferences, correlational analyses were conducted. The primary units of analysis were the mean valence and arousal ratings calculated for each of the 32 video stimuli. Following the data aggregation procedure, the human normative data—comprising the subjective ratings of 72 participants—were averaged to establish as a benchmark for each video. Simultaneously, the MLLM's responses generated across 72 demographic-specific iterations were averaged to derive a representative model-based affective score for each stimulus.

The Pearson correlation was employed as a parametric measure to assess the linear agreement between the human and MLLM datasets. By evaluating linearity, it was possible to determine if the MLLM's internal rating system is mathematically aligned with the human affective space defined in the CAAV database.

In addition to linear alignment, the Spearman rank-order correlation was utilized to evaluate the monotonic relationship between the two datasets. By comparing the relative ranking of the 32 videos—ordered from lowest to highest based on their valence and arousal scores—this non-parametric analysis determines whether the MLLM can successfully preserve the hierarchical order of emotional stimuli as perceived by humans.

*4.2.1 Comprehensive Correlation Analysis: Overall Human vs. MLLM.* The comprehensive correlation analysis evaluated the alignment between the mean affective ratings of the human participants and the MLLM's zero-shot inferences for the 32 video stimuli. The analysis was conducted across the dimensions of valence and arousal using both Pearson ($r$) and Spearman ($\rho$) coefficients.All reported correlations achieved statistical significance at the $p < 0.01$ level. These comprehensive correlation metrics are summarized in Table 1.

- Valence: For the valence dimension, the Pearson correlation coefficient was $r = 0.605$ ($p < 0.01$), and the Spearman rank-order correlation was $\rho = 0.633$ ($p < 0.01$).
- Arousal: For the arousal dimension, the Pearson correlation coefficient was $r = 0.624$ ($p < 0.01$), and the Spearman rank-order correlation was $\rho = 0.568$ ($p < 0.01$).

These results provide the numerical basis for the global agreement between the human normative data and the model-generated ratings, confirming that both the linear relationship and the monotonic rank-order consistency are statistically significant across both affective dimensions.

*4.2.2 Subgroup Correlation Analysis: Age Group-Based Analysis.* A subgroup analysis was performed by stratifying the dataset into young and older adult groups to determine if the MLLM effectively simulates age-related variations in affective perception. All reported correlations for both subgroups achieved statistical significance at the $p < 0.01$ level. The correlation metrics of these subgroups are summarized in Table2.

**Valence Analysis.** In the valence dimension, the correlation coefficients indicated a higher degree of alignment for the older adult subgroup compared to the young adult subgroup.

- Young Adult Subgroup: The Pearson correlation was $r = 0.512$ ($p < 0.01$), and the Spearman rank-order correlation was $\rho = 0.497$ ($p < 0.01$).
- Older Adult Subgroup: The older adult persona exhibited stronger alignment with normative data, yielding a Pearson correlation of $r = 0.635$ ($p < 0.01$) and a Spearman correlation of $\rho = 0.718$ ($p < 0.01$).

Notably, the Spearman coefficient for the older adult subgroup ($\rho = 0.718$) was 0.221 higher than that of the young adult subgroup ($\rho = 0.497$), indicating a more consistent rank-order agreement within the older demographic.

**Arousal Analysis.** A similar trend of demographic variation was observed in the arousal dimension, with the older adult subgroup demonstrating higher correlation metrics.

- Young Adult Subgroup: The analysis resulted in a Pearson correlation of $r = 0.488$ ($p < .001$) and a Spearman correlation of $\rho = 0.466$ ($p < .001$).
- Older Adult Subgroup: The coefficients were markedly higher for this group, with a Pearson correlation of $r = 0.677$ ($p < .001$) and a Spearman correlation of $\rho = 0.654$ ($p < .001$).

A comparison between the groups shows that the Pearson correlation for the older adult subgroup ($r = 0.677$) exceeded that of the young adult subgroup ($r = 0.488$) by 0.189, and the Spearman correlation was 0.188 higher.

The results demonstrate that while the MLLM shows statistically significant alignment with both age groups, the strength of both linear agreement and rank-order consistency is consistently higher when the model evaluates stimuli through the older adult persona.

**Table 1: Comparison of MLLM alignment with human emotion on valence and arousal dimension**

| Metric | Pearson ($r$) | Spearman ($\rho$) |
|---|---|---|
| Valence | 0.605 | 0.633 |
| Arousal | 0.624 | 0.568 |

Note: All correlations are statistically significant ($p < 0.01$).

**Table 2: Comparison of MLLM Alignment across Demographic Subgroups**

| Metric | Pearson ($r$) | Spearman ($\rho$) |
|---|---|---|
| *Valence* | | |
| Young Adult Group | 0.512 | 0.497 |
| Older Adult Group | 0.635 | 0.718 |
| *Arousal* | | |
| Young Adult Group | 0.488 | 0.466 |
| Older Adult Group | 0.677 | 0.654 |

Note: All correlations are statistically significant at $p < 0.01$ level.

# 5 Discussion

## 5.1 MLLM-as-a-Judge for Affective Video

*5.1.1 Alignment with Human Judgment.* Our findings demonstrate the potential and current limitations of using MLLMs as evaluators for affective video content beyond static image stimuli [14]. Regarding RQ1, we observed a mid-to-strong positive correlation between aggregated MLLM responses and human response in both valence and arousal dimensions. By aggregating 36 MLLM calls per video, we successfully mitigated the inherent randomness of individual model outputs, resulting in a robust alignment with human perception across different video stilumi. Consequently, we accept RQ1, confirming that MLLMs can effectively mirror collective human emotional judgment when utilizing pictorial scales like SAM.

An analysis of the distribution reveals that the 32 video stimuli were predominantly concentrated in the mid-to-low valence range (negative affect), a trend reflected in both human and MLLM scores. In contrast, arousal scores were widely distributed across the 1–9 scale for both groups. This suggests that while valence—the binary of pleasantness—tends to be more consistent, arousal—the intensity of emotional activation—exhibits higher variance across different contexts [22, 23], which the MLLM was able to capture to a significant degree.

However, a discrepancy emerged in the Standard Deviation (SD) analysis. While mean scores correlated strongly, the variance (SD) of MLLM responses was notably narrower than that of humans. Although our persona prompting strategy encouraged diverse outputs, the MLLM still struggled to replicate the full breadth of human emotional divergence. This aligns with existing literature suggesting that MLLMs, as probabilistic models, tend to gravitate toward *most likely* or average scores. To more accurately simulate the wide distribution of human responses, more sophisticated prompting techniques [19]or systemic architectural adjustments may be required.

*5.1.2 Representation of Demographic Subgroups.* Regarding RQ2, we examined the model's ability to reflect specific age groups (Young vs. Aged). Interestingly, the correlation was consistently higher in the Aged group across both dimensions, peaking at a Spearman correlation of 0.718 for arousal. The Young group showed a relatively lower but still moderate correlation, with valence at 0.466. While these results indicate that MLLMs can differentiate between subgroups to some extent, we cannot fully accept RQ2 at this stage [3]. The varying performance across demographics suggests that MLLMs may require stronger prompting or one-shot examples of subgroup-specific tendencies to truly internalize and reflect the nuanced emotional profiles of diverse human populations.

## 5.2 Qualitative Content Analysis

To better understand the reasoning behind MLLM judgments, we performed a content analysis on specific cases where the model showed unique patterns or discrepancies.

*5.2.1 Zero Variance in Valence: Ethics and the Absence of Humans.* Zero Variance in Valence: Ethics and the Absence of Humans In few (3 out of all 64) cases, the MLLM produced a result of standard deviation (SD) being 0, meaning all 36 simulated personas provided the exact same score. This occurred when simulating valence scores in two distinct scenarios. First, in videos with strong moral or safety-related themes—such as a video featuring a swastika (Fig. 3.A) or an assault scene (Fig. 3.B)—the MLLM consistently rated them with the lowest valence score of 1 in 1-9 scale. While these responses align with current ethical AI development and safety guidelines, they represent a form of AI hard-coding that lacks the subtle intensity variance often found in human data. Second, we observed a similar zero-variance pattern in mundane, non-human scenes, typically in a video of an umbrella closing. All personas uniformly assigned a mid-valence score of 5, noting that the action "elicits no strong positive or negative emotional response (Fig. 3.C)." This

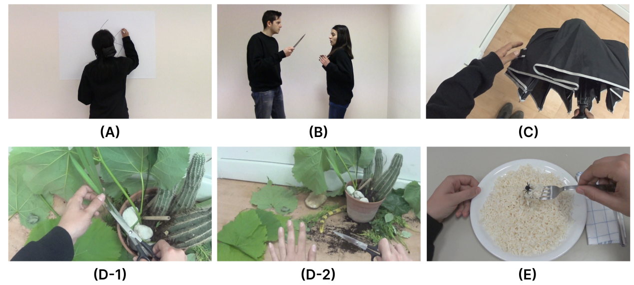


**Figure 3: This figure shows illustrative frames from video stimuli that elicited distinct MLLM judgment behaviors. (A) Swastika: A low-valence, zero-variance scene triggering ethical AI. (B) Assault with Knife: Another low-valence, zero-variance scenario due to a moral taboo. (C) Closing Umbrella: A mundane, non-human action resulting in a neutral mid-valence. (D) Plot Twist in the End: (D-1) Peaceful Pruning is abruptly interrupted by (D-2) Appearance of a Snake, causing inconsistent arousal judgments across MLLM calls. (E) Spider in Food: A scene where despite identifying the spider as fake, the model predicts high human disgust and shock.**

suggests that while MLLMs are sensitive to social taboos [2, 17], they may struggle to project diverse emotional nuances onto scenes lacking human actors [40], defaulting to a neutral baseline when no social or emotional cues are present to trigger a specific persona's perspective.

*5.2.2 Inconsistent Understanding of Plot Twist at the End.* We observed instances where MLLMs struggled with videos containing sudden plot twists at the very end. For example, in a video where a calm scene of pruning plants (Fig. 3.D-1) is suddenly interrupted by a snake (Fig. 3.D-2), most of the some persona instances captured the climax, assigning an arousal score of 9 by noting that the "unexpected appearance of a realistic-looking snake would cause a significant startle." In contrast, few instances assigned an arousal score of 1, focusing only on the "mundane and non-stimulating activity" of pruning. This inconsistency suggests that MLLMs may have an unstable temporal attention span window [10]; depending on the specific call, the model may fail to weigh the final few seconds of a video, leading to an affective judgment that misses the emotional impact of the content.

*5.2.3 Reading Context of Prank and Fakeness.* MLLMs demonstrated a high capacity for understanding the subtle prank atmosphere [36], identifying when a scene was associated with a staged joke or featured fake props. In a scene where a man smiles while handling a bottle labeled as poison, the MLLM did not simply categorize the video as a threat. Instead, it identified the smiling facial expression as a signal of a staged prank. Yet, even in such cases where it explicitly identified that a spider (Fig. 3.E) or snake was a toy, it reasoned that a human observer would still experience intense disgust or shock due to the context of the situation. This indicates that MLLMs do not merely react to visual pixels; they can adopt a Theory of Mind [12] approach, predicting how a human would feel in a given situation despite knowing the situation is staged [38].

This ability to read the line and understand the atmosphere of a prank shows that MLLMs can simulate emotional empathy beyond simple context recognition, predicting how humans will feel despite the detected visual cues [41].

## 5.3 Limitation and Future Works

This study evaluated the efficacy of MLLMs as simulators for subjective emotional perception in video contexts. Several avenues for future research remain in two ways: individual assessment normalization and contextual feature granularity. While our persona prompting encouraged diversity, the model still exhibited a narrower variance (SD) compared to humans. Future work should investigate methods to reflect and normalize individual assessment fluctuations, perhaps by incorporating more granular psychological profiles into the prompting strategy to better simulate the long tail of human emotional response. Also, a limitation of our current stimuli set was the concentration in mid-to-low valence (negative affect) videos. To validate our findings across the full emotional spectrum, we plan to expand the dataset with 58 additional videos that cover high-valence (positive affect) scenarios, ensuring the model's reliability in pleasant or exciting contexts.

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